HAMS App

Wialon Integration Reference

For verification by the P99L hardware partner

Version 1.0

2026-05-25

1. What the app does

- Android app for FFB harvesters; replaces the P99L hardware tracker for cut counting.
- Worker presses **+** per FFB cut — the app stores GPS, battery, and timestamp locally.
- Batched events are pushed to the same Wialon platform as P99L when validated Wi-Fi appears.
- Uplink only — the app does not receive commands or replies from Wialon.

2. UI workflow

3. Wialon server target

Item	Value
IPS host	`185.213.1.24` (DNS: `nl2.gpsgsm.org`)
IPS port	`20332`
Protocol	Wialon IPS v2.x, v1.1 frame variant (text frames over TCP) Spec: https://help.wialon.com/en/wialon-hosting/user-guide/hardware/connecting-hardware#ivedesignedmyowngpsdevicewhichcommunicationprotocolshouldiimplement
REST API host (read-side only)	`https://hst-api.wialon.eu`
Web UI for verification	`https://pro.navi-agnostics.com`

No password used. Login frame: #L#<unique_id>;NA\r\n → expect #AL#1.

4. Unit configuration

The Android app is currently set up against the existing test unit. Please access it on the Web UI and verify settings the same way the P99L unit was verified.

Identity

Item	Value
Test unit name	`TEST_HAMS_APP_001`
Test unit ID	`601602811`
Unique ID	`HAMS_TEST_001`
Device type	Wialon IPS (ID `600002235`)
Password	*(blank)*

Required Advanced-tab filters

Filter	Required value
Maximum interval between messages (s)	`0`
Average speed between messages	`0`
Distance between coordinates	`0`
Message validity filtration	Off

Critical: any non-zero or "on" value here causes Wialon to silently drop messages — no #AD# failure, no Messages-tab entry. Same constraint as the P99L unit.

5. Sensors on the unit

Sensor	Parameter	Type	Notes
`FFB_CUT`	`ffb_cut`	Custom	1 = cut event
`battery`	`battery`	Custom (units %)	Phone battery percent
`event_code`	`event_code`	Custom	Optional calibration: 179 / 180 / 35
`work_count`	`work_count`	Counter	Displayed/net count per task

Parameter-naming difference from P99L. The HAMS app uses descriptive parameter names (`ffb_cut`, `battery`, `event_code`, `work_count`) instead of P99L's short single-letter names. Both are valid per IPS v1.1, but Wialon sensors created for HAMS units must use the descriptive names shown above.

6. Data frame — what the app sends

Standard 16-field IPS v1.1 data frame over TCP, one frame per event:

```
#D#DDMMYY;HHMMSS;DDMM.MMMM;N;DDDMM.MMMM;E;speed;course;alt;sats;hdop;inputs;outputs;adc;ibutton;params\r\n
```

Field	App-side value
date / time	UTC at the moment the event was captured.
latitude / longitude	DDMM.MMMM — latitude uses 2-digit degrees, longitude uses 3-digit degrees.
speed	`0` — worker on foot.
course	`0` — no heading. The legacy V5 `course=1` FFB-hack is removed.
altitude	Terrain estimate.
satellites / hdop	From the Android Fused Location provider.
inputs / outputs	`0` / `0` (not applicable for a phone).
adc	Empty.
ibutton	NA
params	Custom params block (next section).

Expected gateway response per frame: #AD#1.

Custom params block (field 16)

```
ffb_cut:1:<0|1>,battery:2:<0.0-100.0>,event_code:1:<179|180|35>,work_count:1:<0..>
```

Param	Type code	Meaning
`ffb_cut`	1 (int)	1 on **+** press, `0` on every other event
`battery`	2 (double)	Phone battery percent at capture
`event_code`	1 (int)	Outbound code — see next section
`work_count`	1 (int)	Displayed/net task count after this event

7. Outbound event codes

Only three codes are ever sent to Wialon.

Code	Meaning	Frequency
179	FFB cut / plus	Per worker **+** press
180	FFB correction / productive minus	Per worker **–** press, only when net count > 0
35	Periodic beacon (heartbeat)	Once per minute while a task is active

App-internal lifecycle and health codes are stored on the device only and never reach Wialon.

Sample

#D#180526;090519;0435.9886;N;10104.6871;E;0;0;10;8;20;0;0;;NA;ffb_cut:1:1,battery:2:87,event_code:1:179,work_count:1:1\r\n

Server reply: #AD#1\r\n

Frame field	Came from
`180526;090519`	Time 2026-05-18 09:05:19 UTC
`0435.9886;N`	Lat 4.59981 → 4° + 0.59981 × 60′
`10104.6871;E`	Lon 101.078118 → 101° + 0.078118 × 60′
`0;0;10`	Speed / course / altitude
`8`	Satellites — not shown in the Messages tab; placeholder. Open Raw data to read the real value.
`20`	HDOP
`0;0;;NA`	inputs / outputs / adc (empty) / ibutton
`ffb_cut:1:1,battery:2:87,event_code:1:179,work_count:1:1`	Params block

8. Push trigger and batching

Each event is captured and stored locally. The app pushes to Wialon when validated Wi-Fi becomes available, in the following sequence.

#	Step	Detail
1	Worker presses + on the count screen	Event written to local SQLite with timestamp, GPS, battery, event_code=179, and work_count after the press.
2	Event stored locally	Status: pending. App keeps storing events offline indefinitely (capacity ~30 days).
3	Push trigger fires	Either (a) a validated unmetered Wi-Fi network connects, or (b) the user long-

#	Step	Detail
		presses the manual push button for 3 seconds. Mobile data is never used.
4	App opens TCP socket	Destination 185.213.1.24:20332. Socket is opened fresh per batch — no long-lived connections.
5	Login	App sends #L#<unique_id>;NA and waits for #AL#1 . On #AL#0 (rejected) the socket is closed and push stops; on timeout the app retries with backoff.
6	Drain a batch	Up to 10 #D# frames per TCP session, ~75 ms between frames. Order: oldest pending event first.
7	Per-frame ack	App reads #AD#<code> after each frame. #AD#1 = event marked uploaded. #AD#-1, #AD#15, etc. = event marked rejected (stays on device, not auto-retried).
8	Close and continue	After 10 frames or queue empty, the app closes the socket. If more events remain, it opens a fresh session (new login) and repeats.
9	Heartbeat	While a task is active, the app emits one event_code=35 frame per minute. Pushed with the next batch — never alone.
10	Failure handling	TCP error / login rejection / timeout: socket closed, backoff 10s -> 30s -> 60s -> 120s, retry when Wi-Fi is still up. App never blocks the worker — counting continues regardless of push state.

Rule	Value
Trigger	Validated unmetered Wi-Fi (monitored by a background service).
Manual override	3-second long-press on the push button in the app.
Frames per TCP session	`10`
Inter-frame delay	~75 ms
Login required per session	Yes (#L#…;NA)

The push monitor stays alive while pending events exist — even with the app closed or swiped from recents — so push fires the moment Wi-Fi appears.

9. What the app does NOT send

- No digital `inputs` / `outputs` bits (both = `0`).
- No ADC values.
- No driver / ibutton key (sent as NA).
- No heading on `course` (sent as `0`); the legacy V5 `course=1` FFB-hack is removed.
- No HAMS-internal lifecycle or health event codes — they remain on the device only.
- No data flows back from Wialon to the app — push is uplink only.

10. How to verify

The same test unit (`TEST_HAMS_APP_001`, ID `601602811`) is being used. Please open it on `pro.navi-agnostics.com` and verify the stored messages the same way the P99L unit was verified during its integration — message arrival, geofence resolution, sensor decode, and report runs.

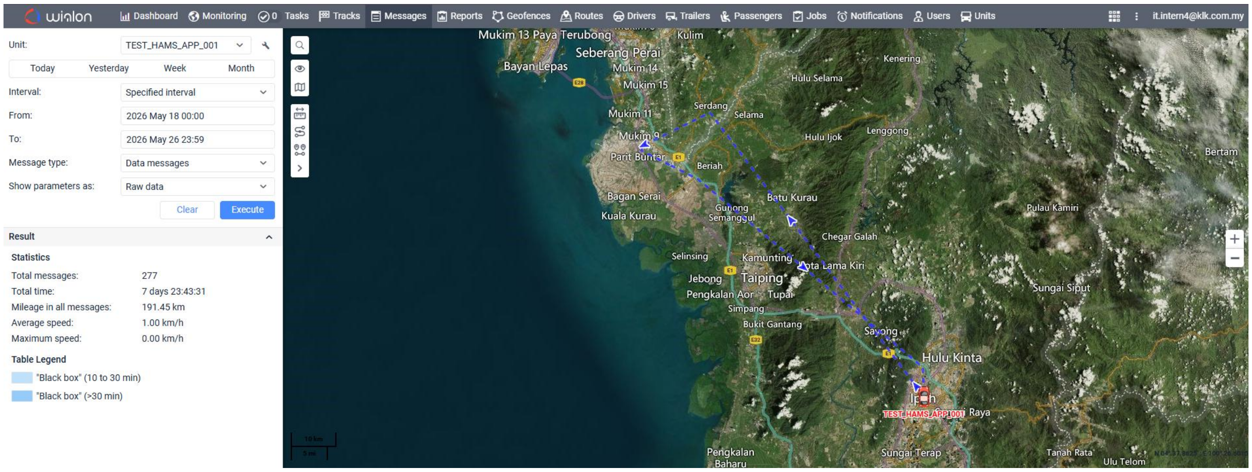

Table Chart

#	Time	Speed, km/h	Coordinates	Altitude, m	Location	Parameters
151	2026-05-21 19:09:49	0	5.05008666667, 100.60884 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=3, I/O=0/0
152	2026-05-21 19:09:54	0	5.04896833333, 100.610293333 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=2, I/O=0/0
153	2026-05-21 19:09:54	0	5.04896833333, 100.610293333 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=1, I/O=0/0
154	2026-05-21 19:09:55	0	5.04879333333, 100.610543333 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=3, I/O=0/0
155	2026-05-21 19:09:57	0	5.048465, 100.61106 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=5, I/O=0/0
156	2026-05-21 19:09:57	0	5.048625, 100.610795 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=4, I/O=0/0
157	2026-05-21 19:10:04	0	5.04746, 100.61293 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=1, I/O=0/0
158	2026-05-21 19:10:04	0	5.04746, 100.61293 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=2, I/O=0/0
159	2026-05-21 19:10:05	0	5.04746, 100.61293 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=3, I/O=0/0
160	2026-05-21 19:10:06	0	5.04733333333, 100.613205 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=4, I/O=0/0
161	2026-05-21 19:10:06	0	5.04721, 100.613485 (0)	10	Lebuhraya Utara–selatan, Mukim Beriah, Kerian, Perak, Malaysia	hdop=0.8, ffb_cut=1, battery=58, event_code=179, work_count=5, I/O=0/0
162	2026-05-22 06:15:27	0	4.66083833333, 101.075138333 (0)	10	Laluan Meru Indah G3, Mukim Hulu Kinta, Perak, Malaysia	hdop=9.6, ffb_cut=1, battery=64, event_code=179, work_count=2, I/O=0/0
163	2026-05-22 06:15:27	0	4.66083833333, 101.075138333 (0)	10	Laluan Meru Indah G3, Mukim Hulu Kinta, Perak, Malaysia	hdop=9.6, ffb_cut=1, battery=64, event_code=179, work_count=1, I/O=0/0
164	2026-05-22 06:15:27	0	4.66083833333, 101.075138333 (0)	10	Laluan Meru Indah G3, Mukim Hulu Kinta, Perak, Malaysia	hdop=9.6, ffb_cut=1, battery=64, event_code=179, work_count=3, I/O=0/0
165	2026-05-22 06:15:32	0	4.6608, 101.075171667 (0)	10	Laluan Meru Indah G3, Mukim Hulu Kinta, Perak, Malaysia	hdop=1.5, ffb_cut=1, battery=64, event_code=179, work_count=1, I/O=0/0
166	2026-05-22 06:15:32	0	4.6608, 101.075171667 (0)	10	Laluan Meru Indah G3, Mukim Hulu Kinta, Perak, Malaysia	hdop=1.5, ffb_cut=1, battery=64, event_code=179, work_count=2, I/O=0/0

End of reference.